zerokost

love vs life

BioPlatform®

ADVANCED BIOLOGICS DISCOVERY

Target Explorer

Explore Target Networks

Virtual Screening

Filter Millions to Key Hits

Structural Docking

Liquid Handling Robots

AI-Lab Loop

ADMET Intelligence

Lead Optimization

ProteinMPNN Design

Wet-Lab Loop

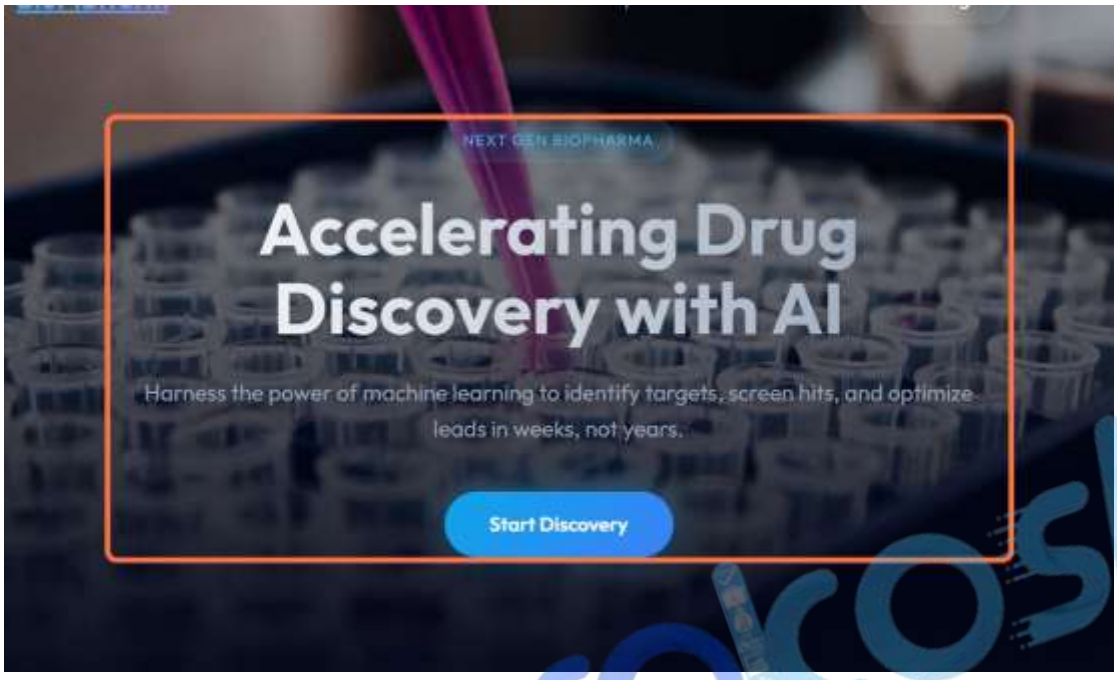
Automated Assays

60% FASTER

40% LOWER COSTS

STEP 1

Click on NEXT GEN BIOPHARMA…



This opens the main portal of the biologics discovery platform where users can perform computational drug discovery workflows including docking, screening, and ADMET analysis.

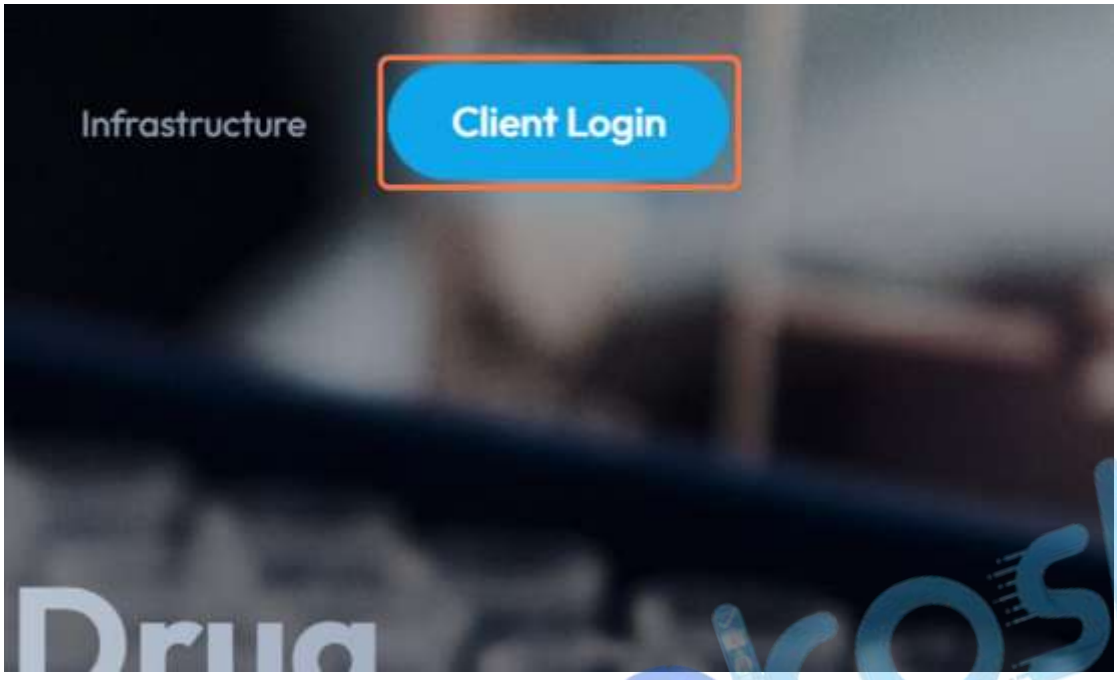
The platform integrates several advanced modules such as:

- Target identification
- Structural docking
- Hit screening
- Virtual compound screening
- AI-based lead optimization
- ADMET prediction
- Wet-lab automation integration

The homepage acts as the **gateway to the full drug discovery pipeline**.

STEP 2

Click on Client Login



This option is used for authorized pharmaceutical researchers, collaborators, or institutional clients who have been granted access to the platform.

The Client Login option is specifically designed for:

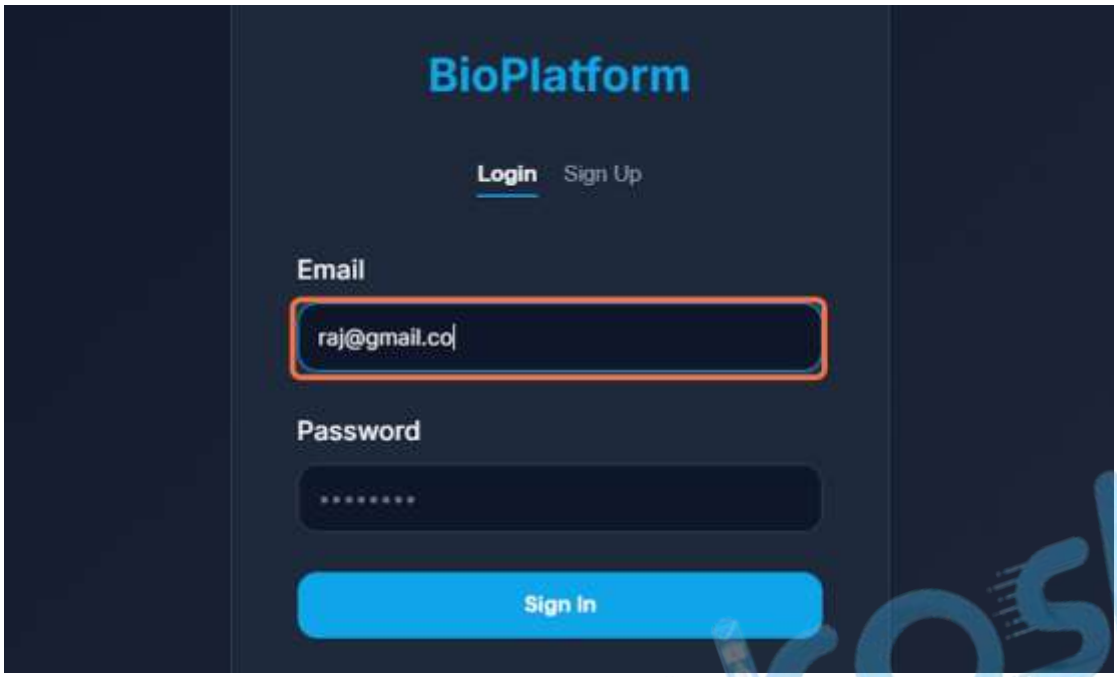
- pharmaceutical researchers
- biotech partners
- academic collaborators
- internal R&D teams

This ensures that only verified users can access confidential project data and computational resources.

If you are a new user and do not yet have credentials, please contact the system administrator or the Zerokost support team to request access.

STEP 3

Type "raj@gmail.com"



The system verifies the user identity based on the registered credentials in the platform database.

The email ID acts as the **primary user identifier** within the platform.

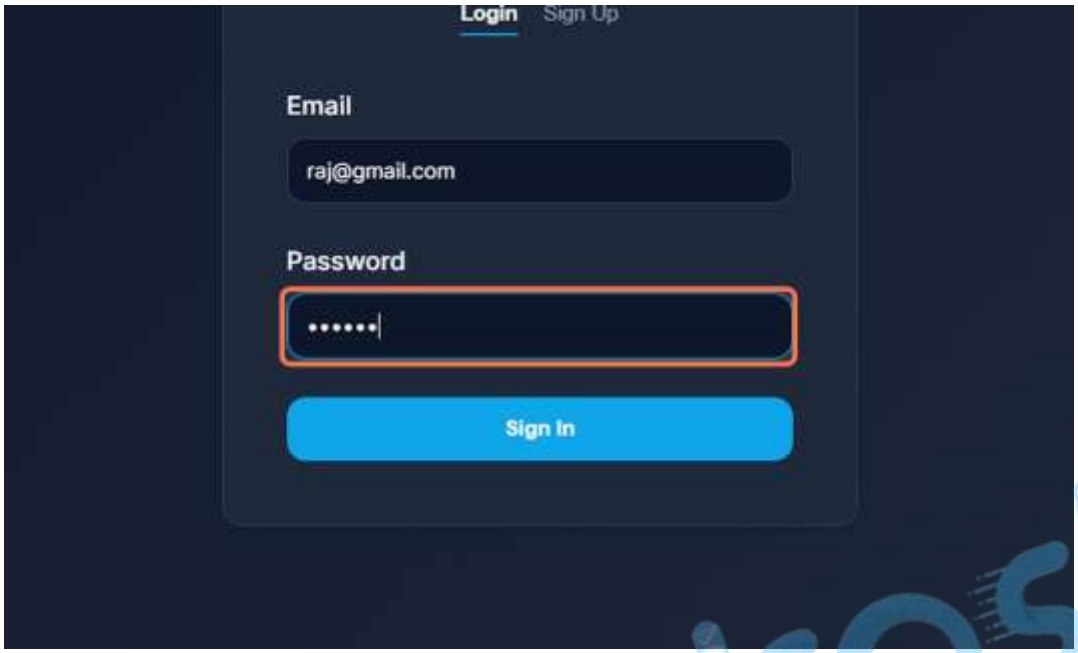
It allows the system to:

- associate your activity with a specific research project
- store docking results
- maintain experiment logs
- save reports and generated molecules

Make sure you enter the **exact email used during account registration**.

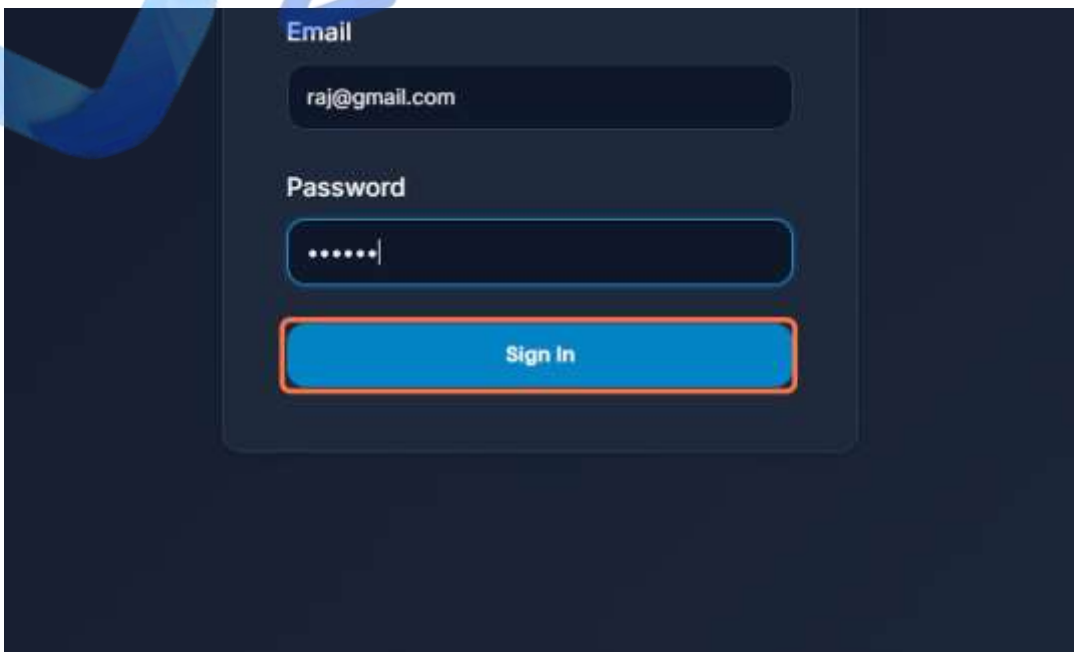
STEP 4

Type password



STEP 5

Click on Sign In



STEP 6

Click on Research Dashboard…

STEP 7:

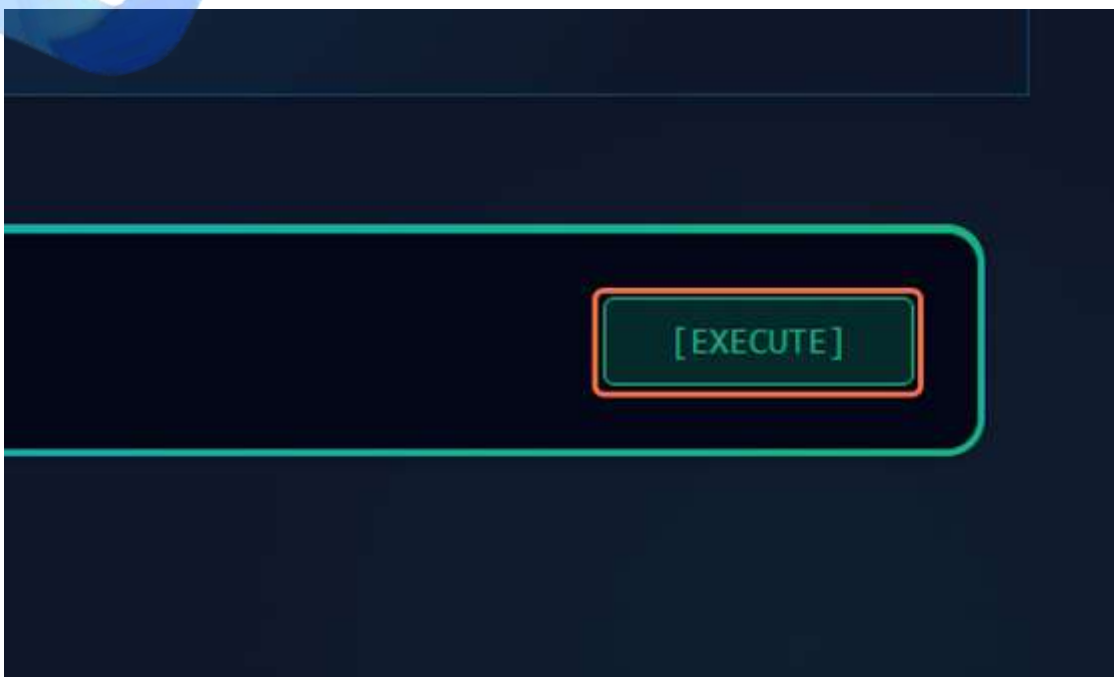
Click on Target Explorer

STEP 8:

Type "5FR9"

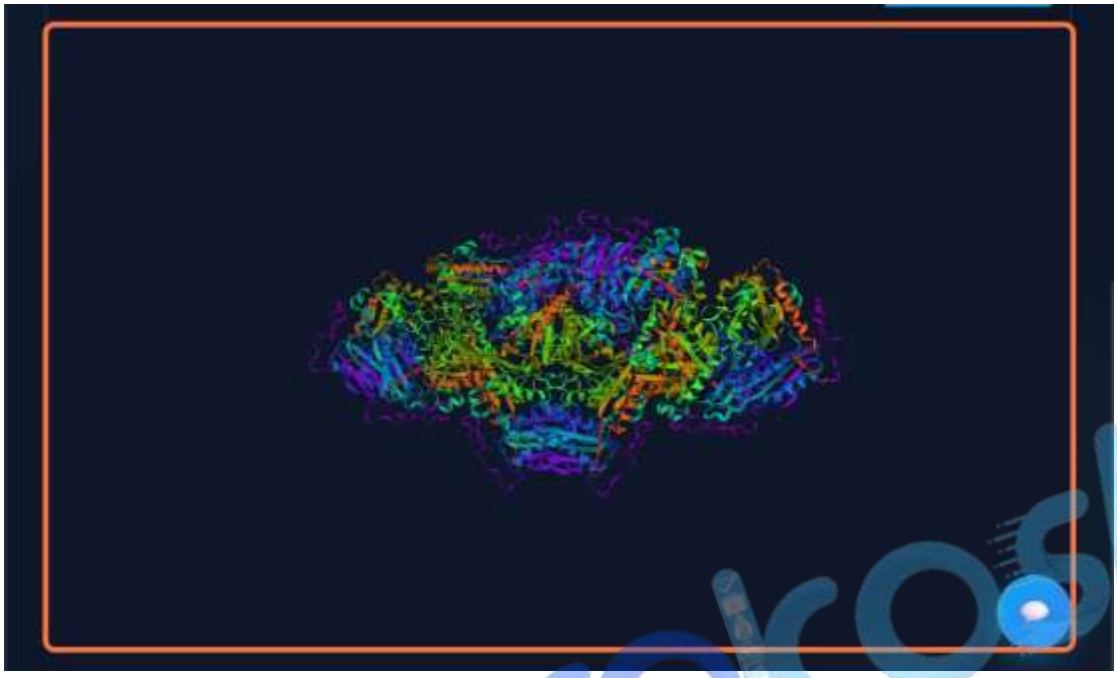
STEP 9:

Click on [EXECUTE]



STEP 10:

Click on highlight



The highlight function identifies **active binding pockets** within the protein structure.

Binding pockets are regions where ligands can bind and interact with amino acid residues.

The system uses structural analysis to detect:

- cavities
- interaction hotspots
- ligand binding residues

These highlighted regions guide the docking simulation.

STEP 11:

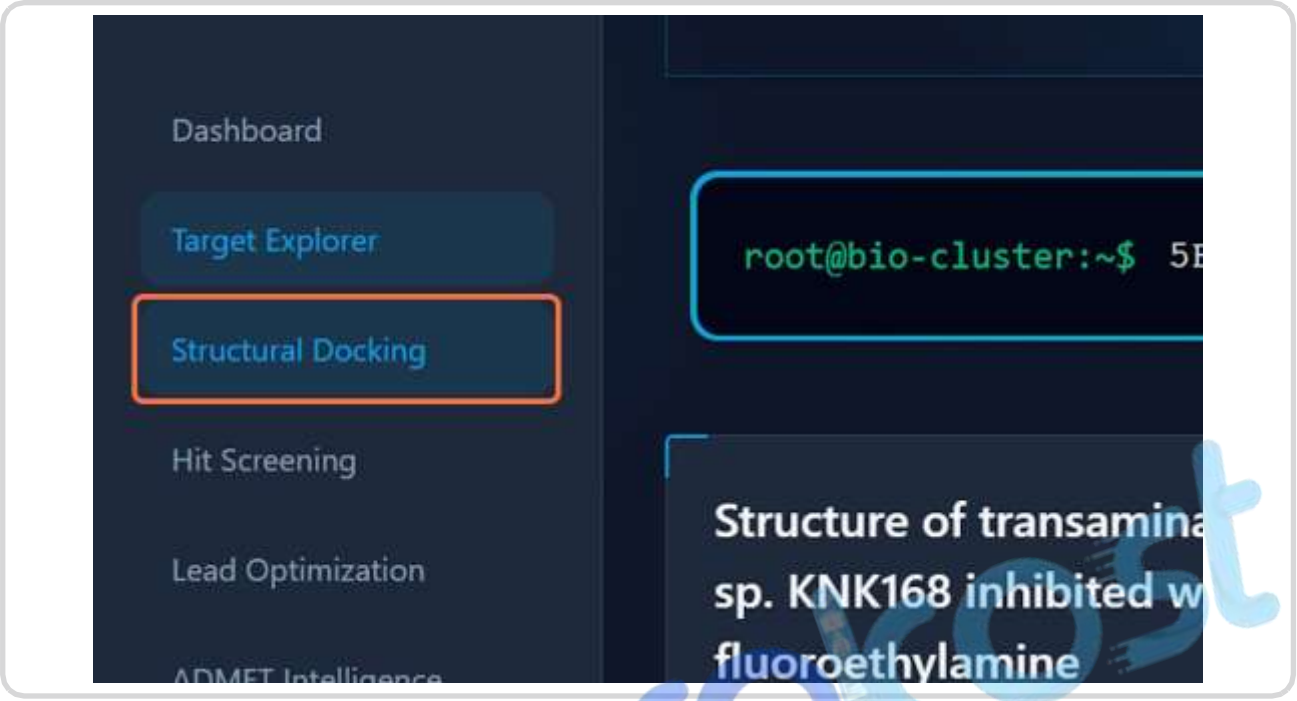
Click on Structure of transaminase ATA-117 arRmut11 from Arthrobacter sp. KNK168 inhibited with 1-(4-Bromophenyl)-2-fluoroethylamine…

This loads the complete protein structure into the visualization environment.

The system now prepares the protein receptor for **molecular docking experiments**.

STEP 12:

Click on Structural Docking



Structural docking predicts how a **ligand molecule binds to a protein target**.

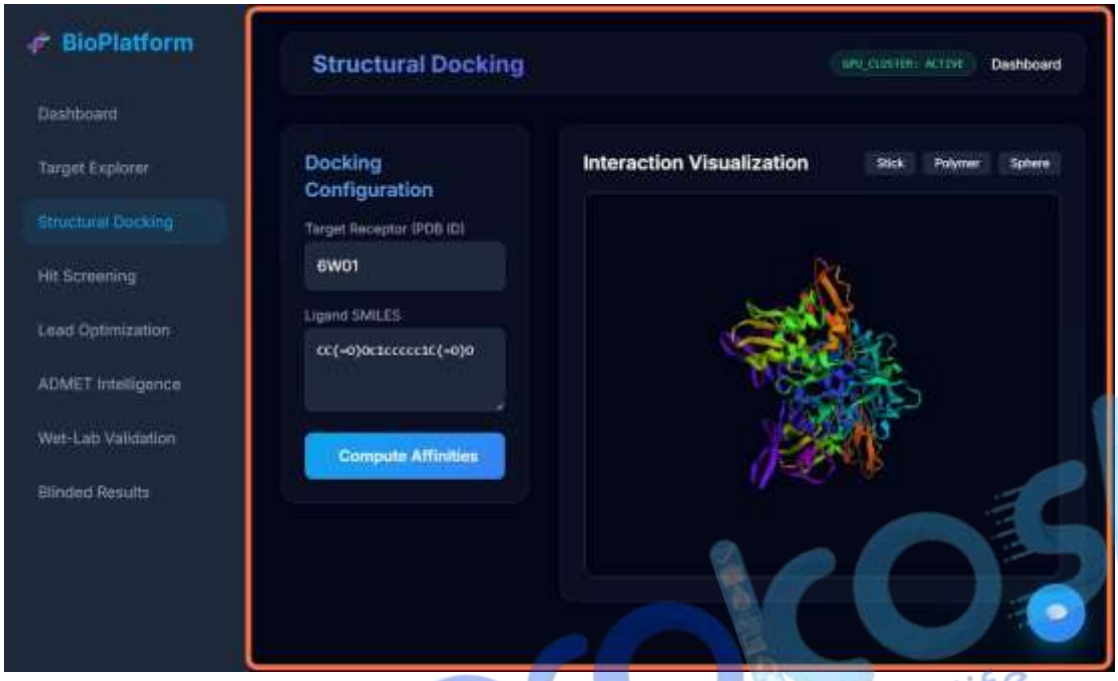
The docking algorithm calculates:

- binding affinity
- interaction energy
- docking pose orientation

These predictions help identify potential therapeutic molecules.

STEP 13:

Click on Structural Docking…



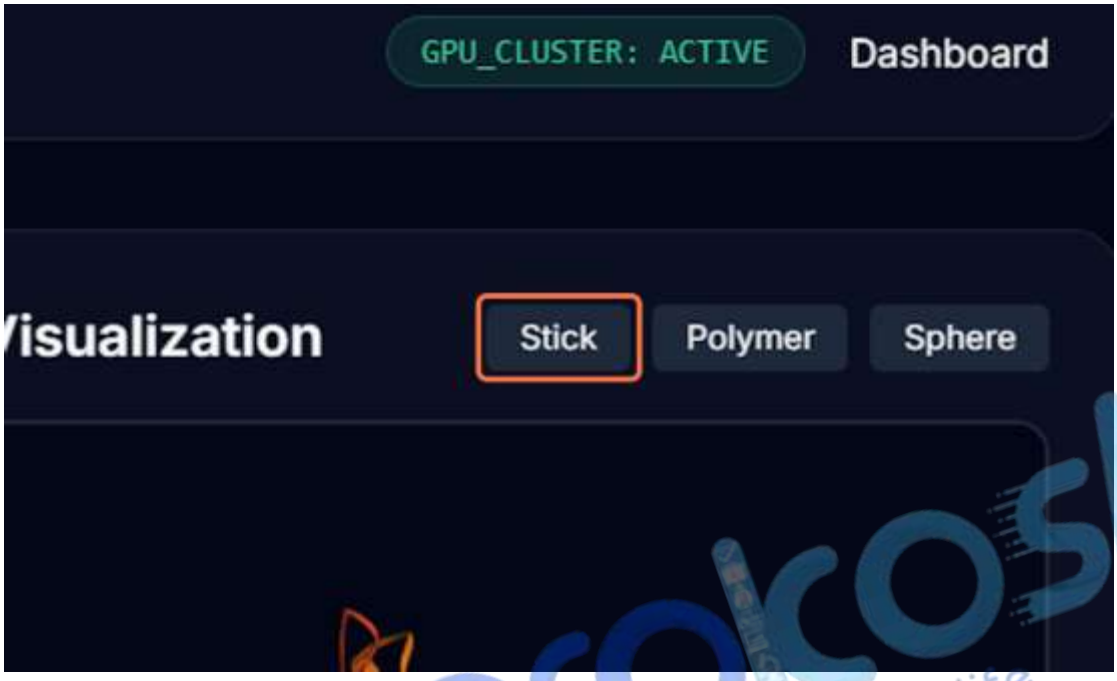
This module initializes the docking environment.

The system prepares the protein structure by:

- removing water molecules
- optimizing hydrogen atoms
- identifying docking grid regions

STEP 14:

Click on Stick

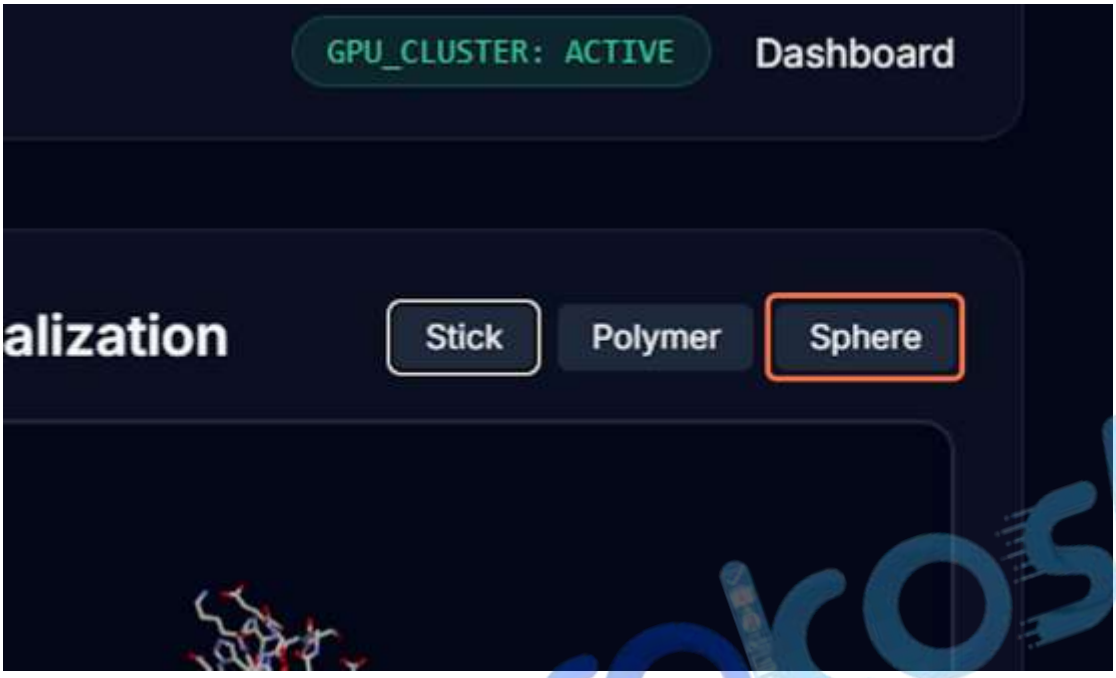


The docking console allows users to configure:

- ligand molecules
- docking parameters
- scoring functions
- simulation options

STEP 15:

Click on Sphere



Stick representation displays molecular bonds clearly.

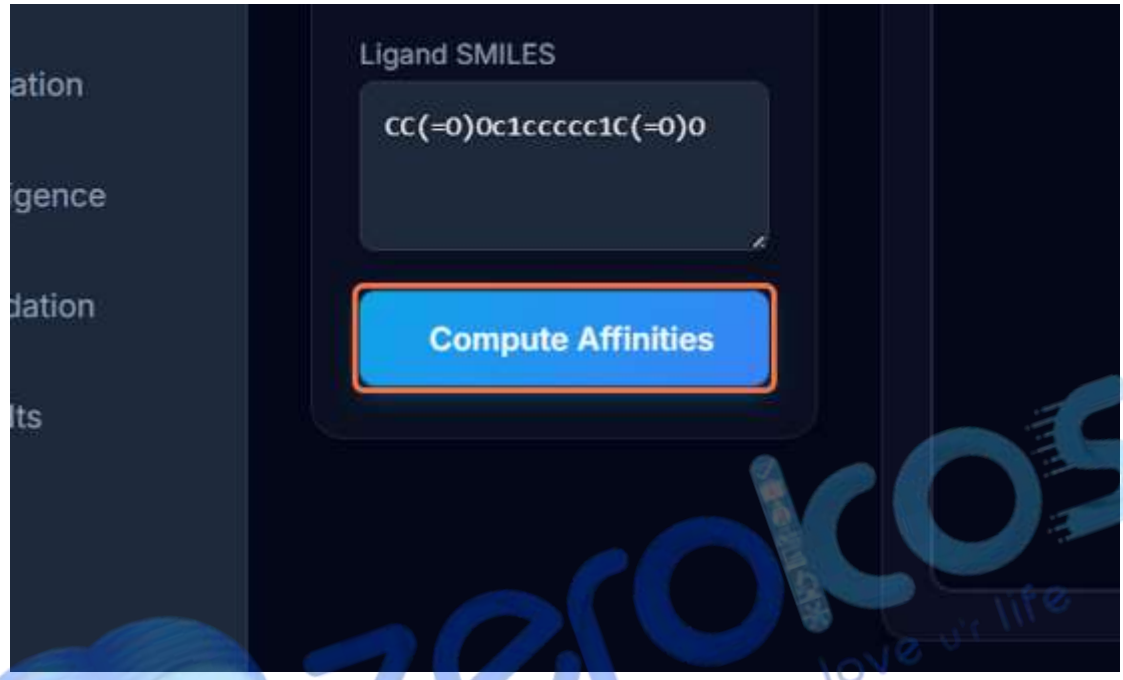
This view is useful for examining:

- ligand orientation
- hydrogen bonds
- molecular interactions

STEP 16:

Click on Compute Affinities

The docking engine calculates the interaction between the ligand and protein.



It estimates:

- binding energy
- interaction strength
- predicted docking pose

The system uses computational chemistry algorithms to simulate ligand-protein binding.

STEP 17:

Click on Job Progress…

The Job Progress panel displays the status of docking simulations.

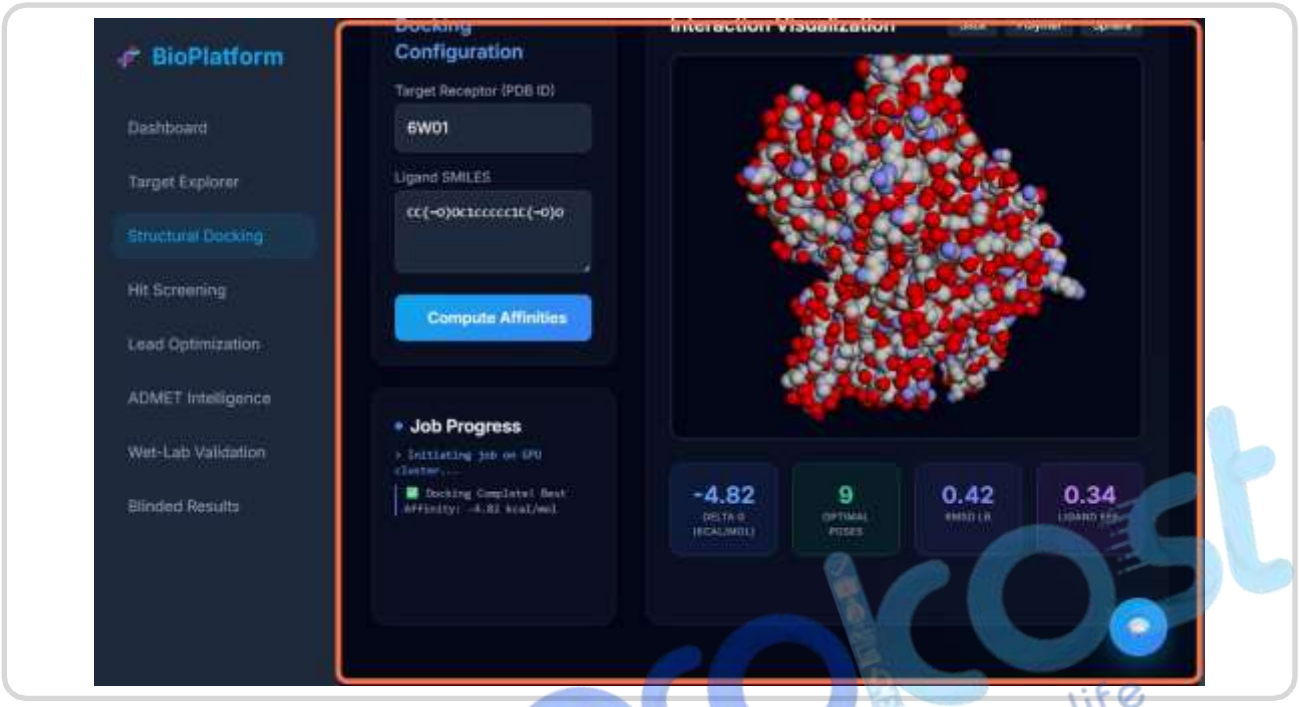
Users can monitor:

- simulation progress
- queued jobs
- completed docking tasks

This ensures transparency and allows users to track long-running calculations.

STEP 18:

Click on Structural Docking…



This module initializes the docking environment.

The system prepares the protein structure by:

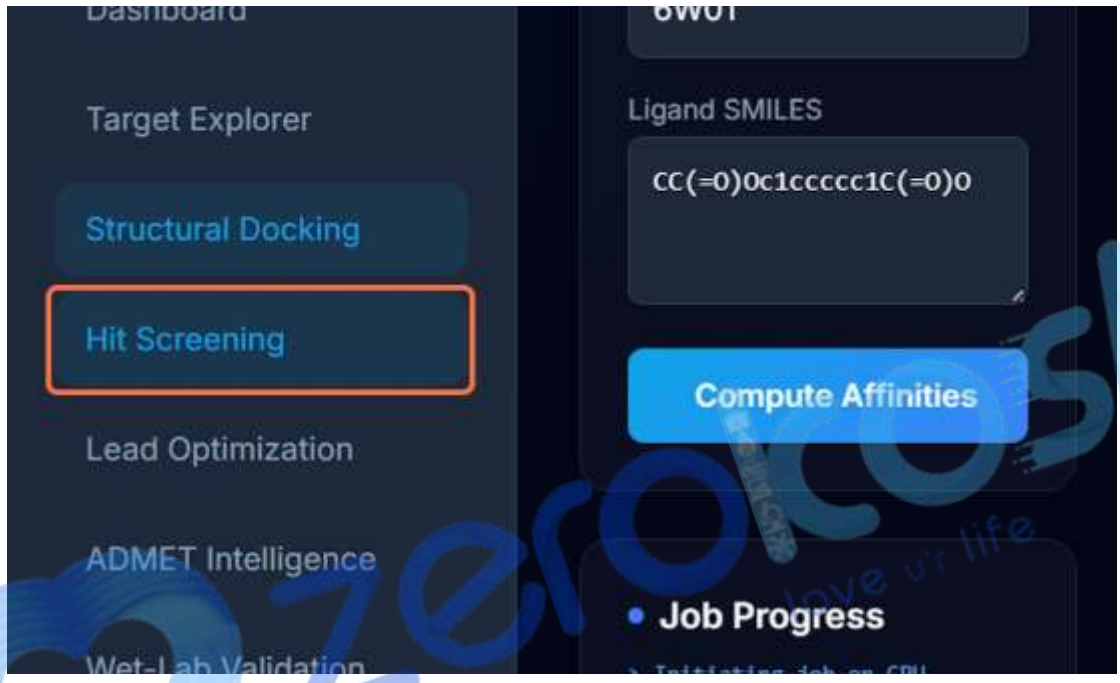
- removing water molecules
- optimizing hydrogen atoms
- identifying docking grid regions

STEP 19:

Click on Hit Screening

Hit Screening identifies **promising compounds (hits)** from experimental datasets.

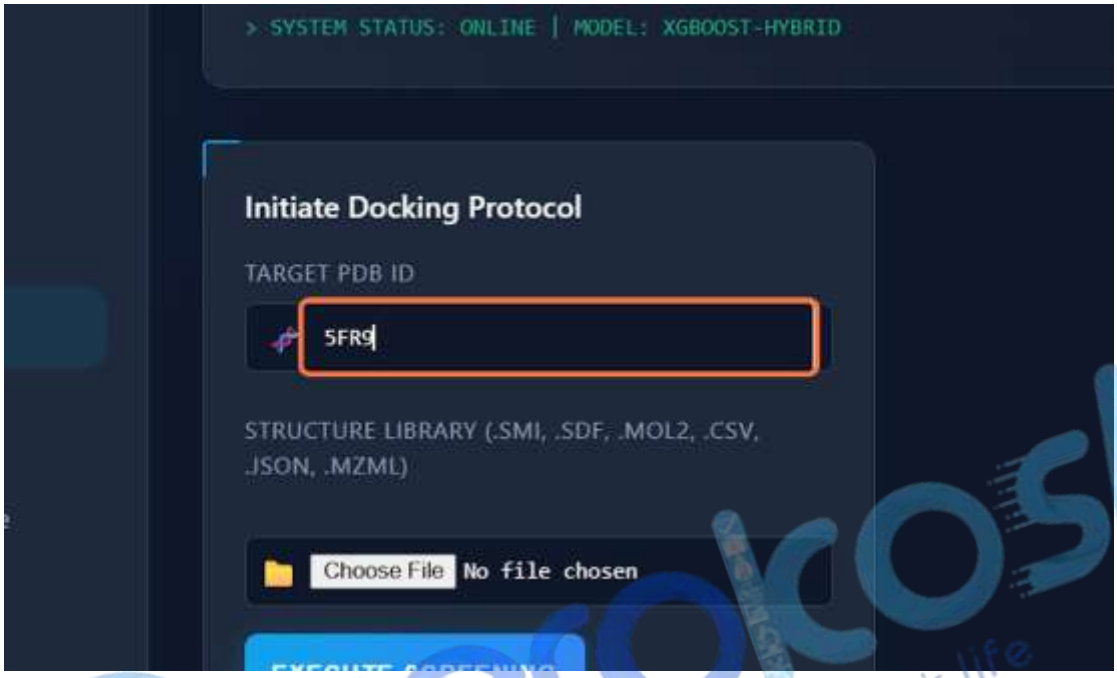
These hits represent molecules that show potential binding activity with the target protein.



STEP 20:

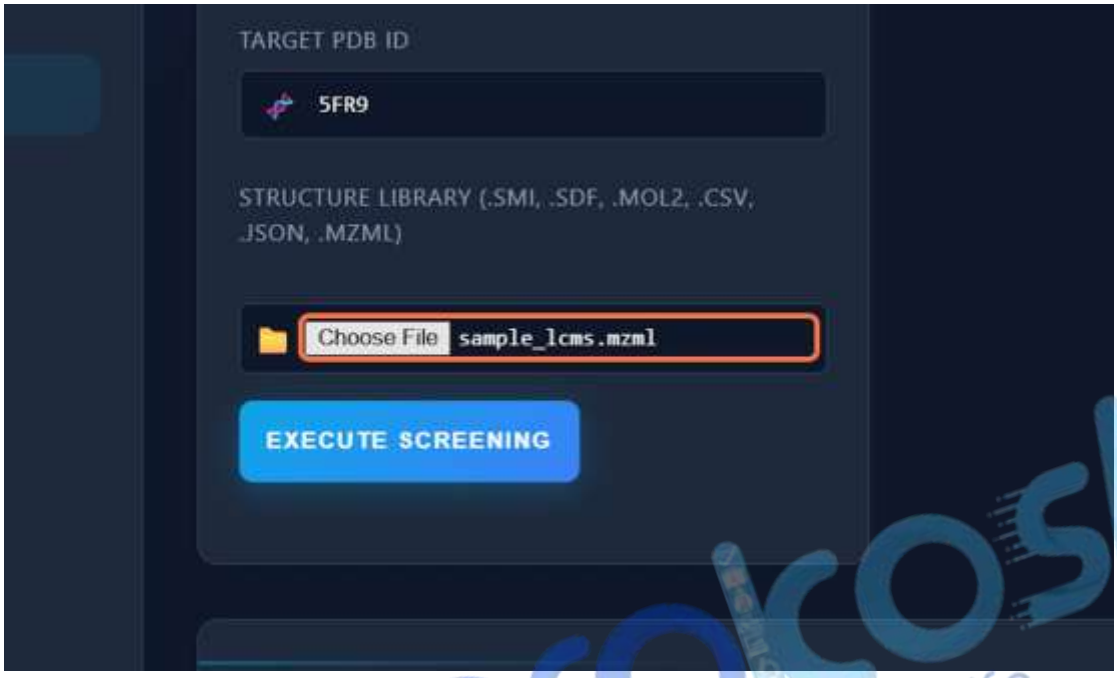
Type "5FR9"

This ensures that screening is performed against the correct protein structure.



STEP 21:

Select "sample_lcms.mzml" from file upload menu



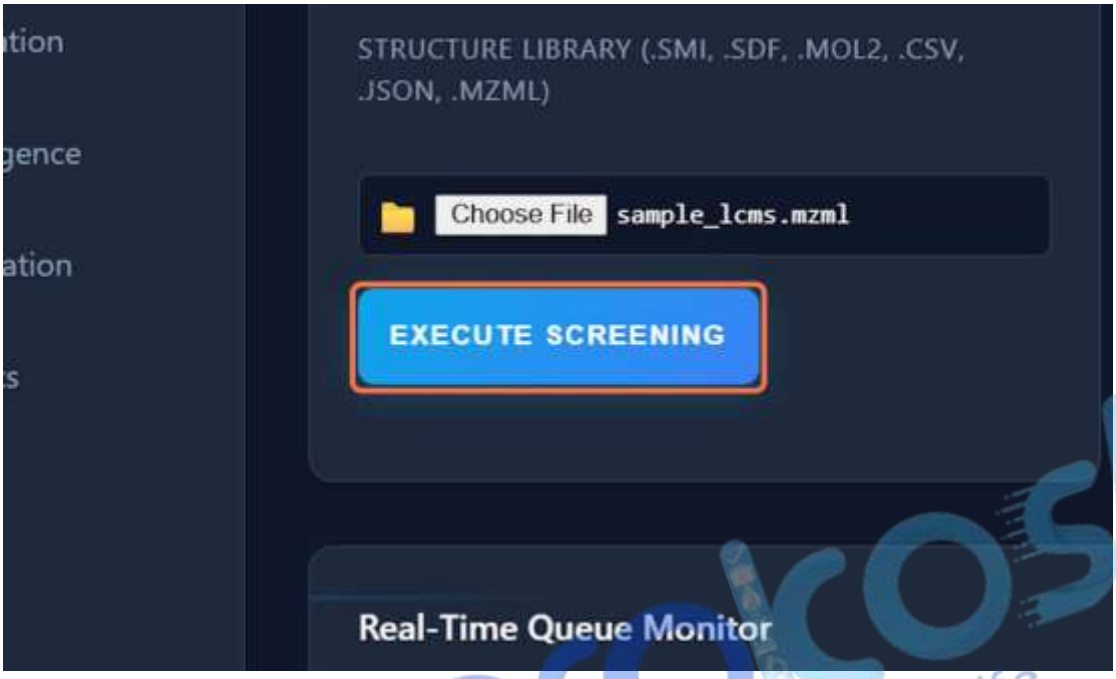
The platform supports experimental mass spectrometry datasets such as LC-MS.

These datasets may contain compound signals detected during biological experiments.

The system processes the dataset and extracts potential candidate molecules.

STEP 22:

Click on EXECUTE SCREENING



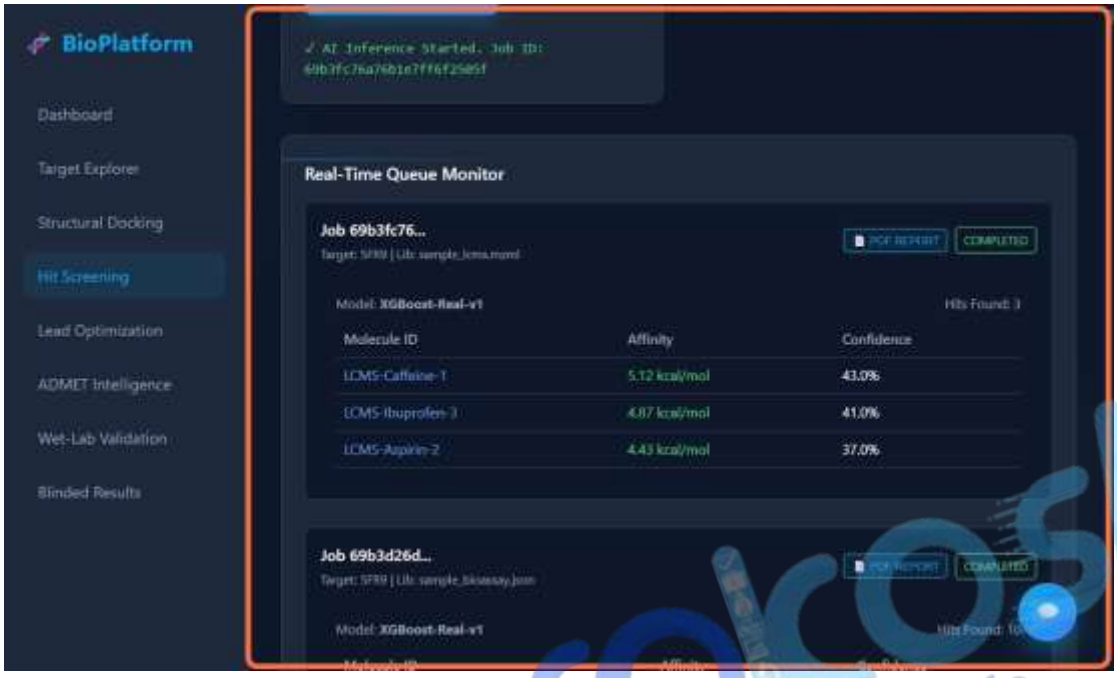
The platform performs several analyses:

- compound identification
- spectral matching
- docking evaluation
- scoring and ranking

The output is a list of **potential hit molecules**.

STEP 23:

Click on Virtual Screening Console…

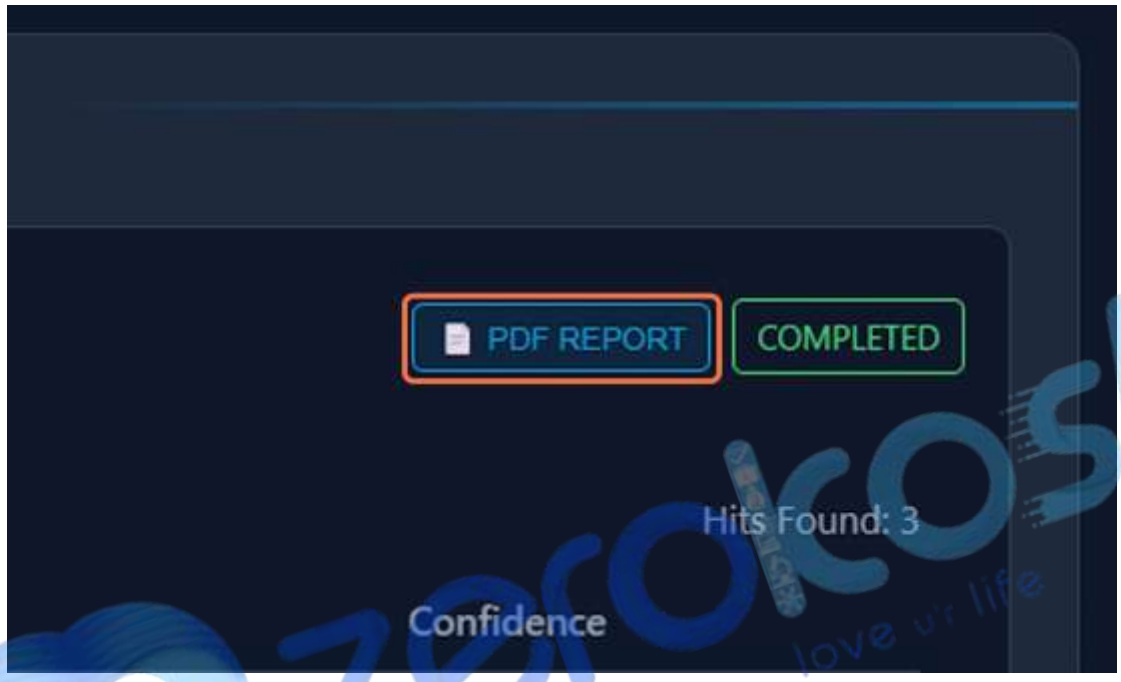


Virtual screening evaluates **thousands or millions of molecules** computationally to identify promising candidates.

STEP 24:

Click on PDF REPORT

This module runs high-throughput computational screening across compound libraries.

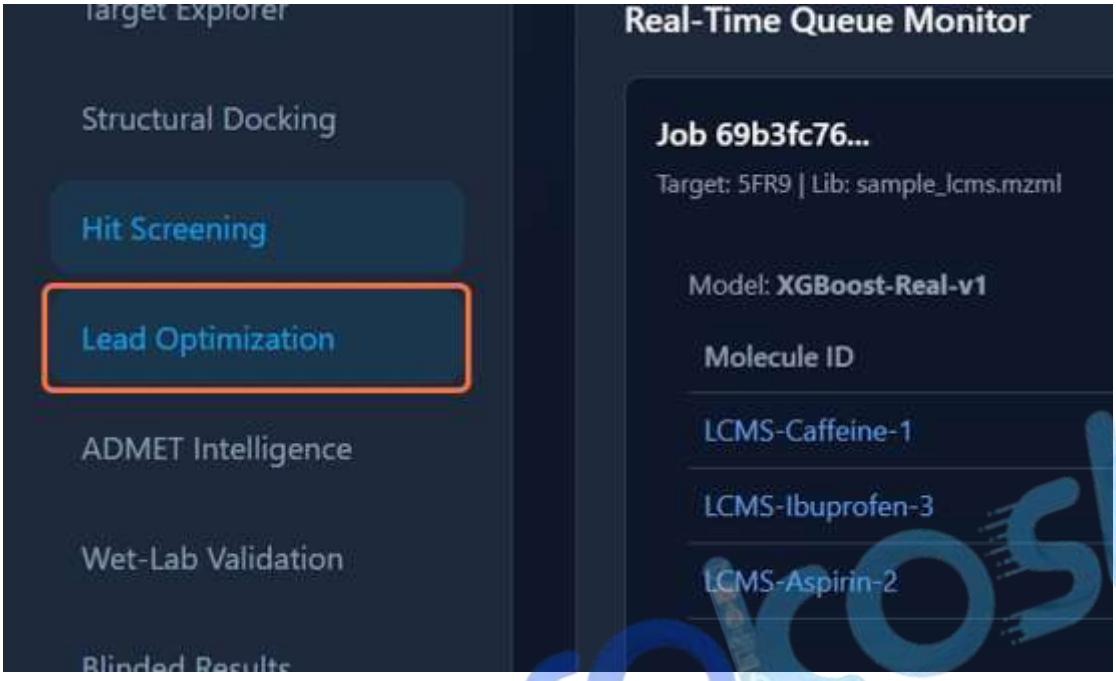


The system evaluates molecules based on:

- docking scores
- structural compatibility
- predicted binding affinity

STEP 25:

Click on Lead Optimization



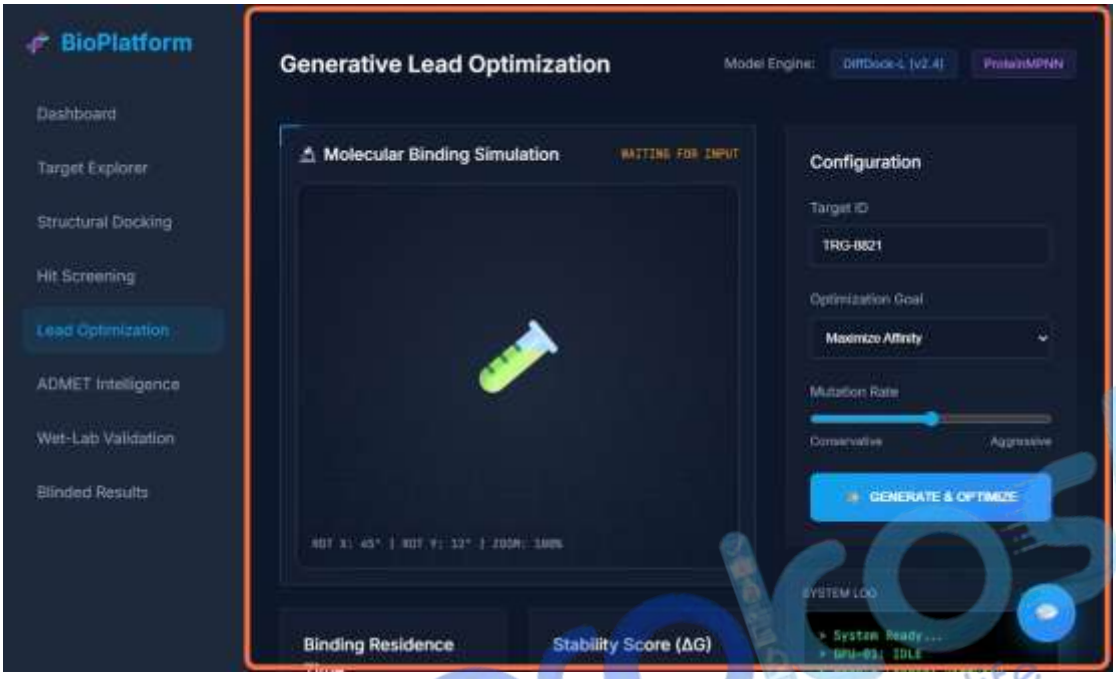
The platform generates a detailed scientific report containing:

- ranked compounds
- docking scores
- interaction maps
- structural visualizations

This report can be shared with researchers or project collaborators.

STEP 26:

Click on Generative Lead Optimization…



Lead Optimization improves candidate molecules using **AI-driven molecular design**.

The goal is to enhance:

- potency
- selectivity
- pharmacokinetics
- safety profile

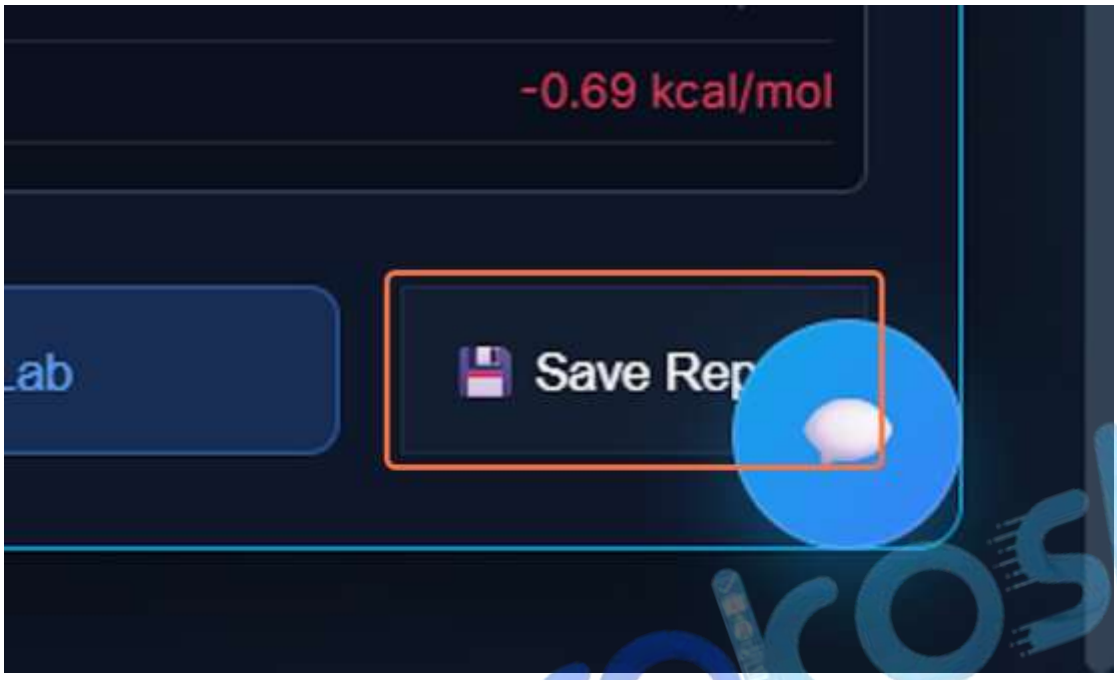
STEP 27:

Click on GENERATE & OPTIMIZE

This module uses generative AI algorithms to propose **new molecular structures** derived from initial hits.

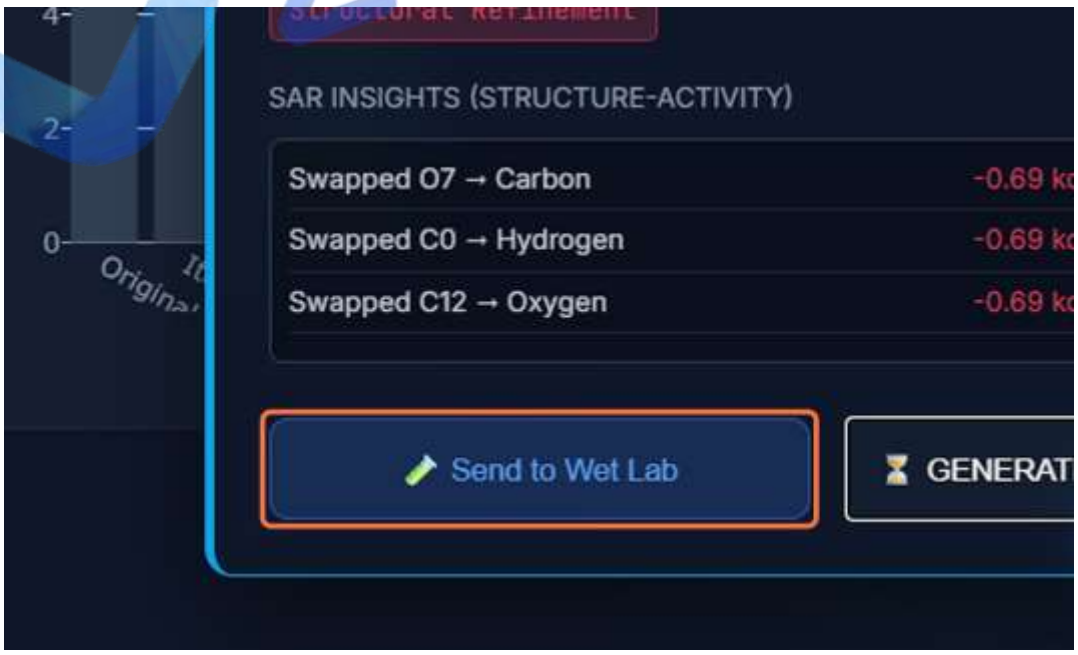
STEP 28:

Click on 💾 Save Report



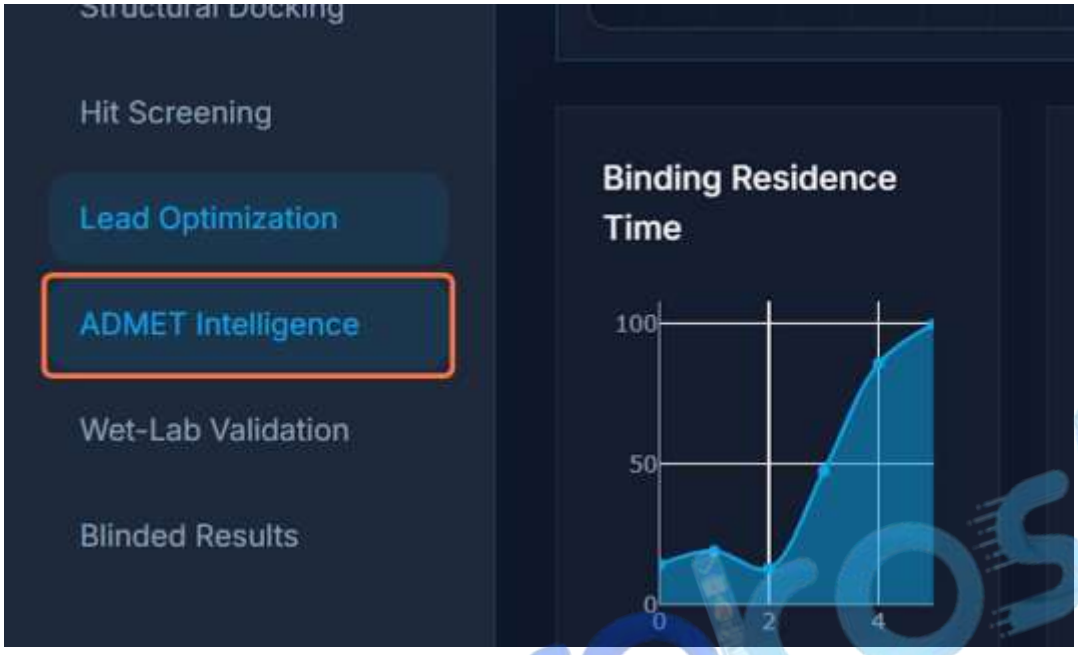
STEP 29:

Click on Send to Wet Lab



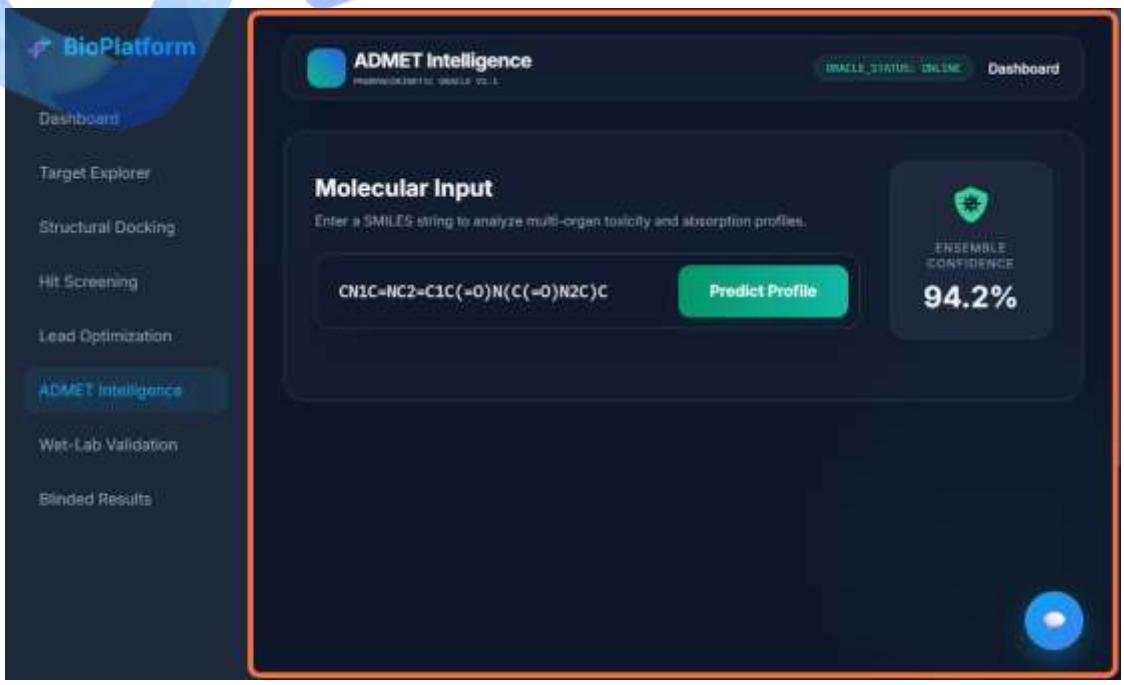
STEP 30

Click on ADMET Intelligence



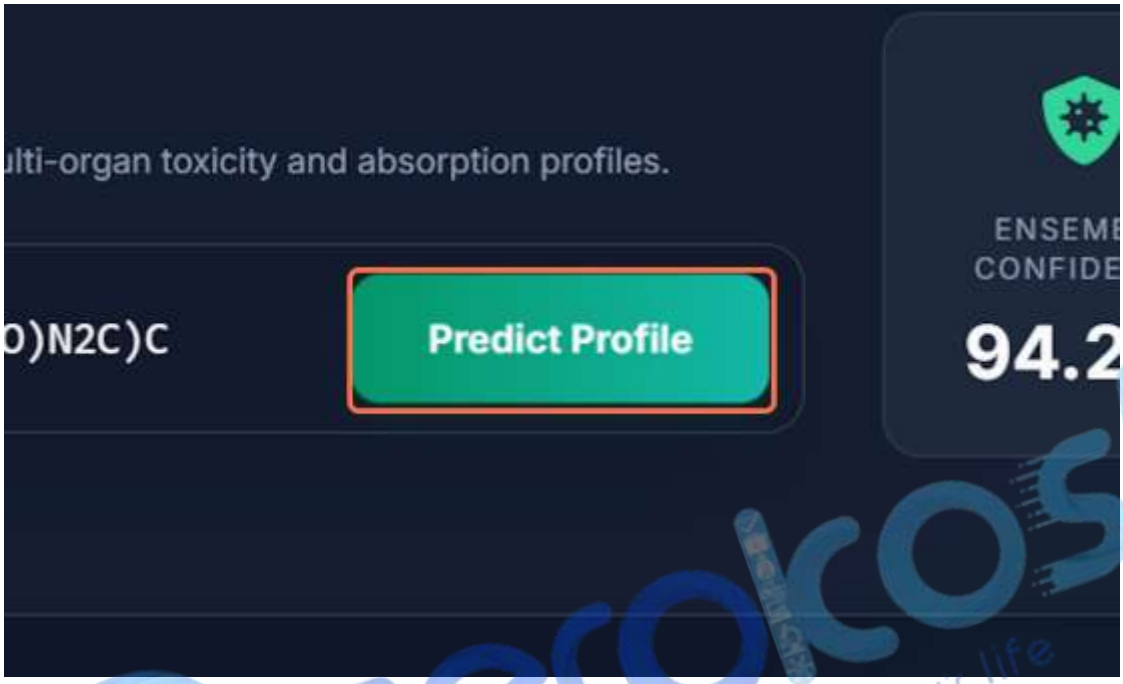
STEP 31

Click on ADMET Intelligence…



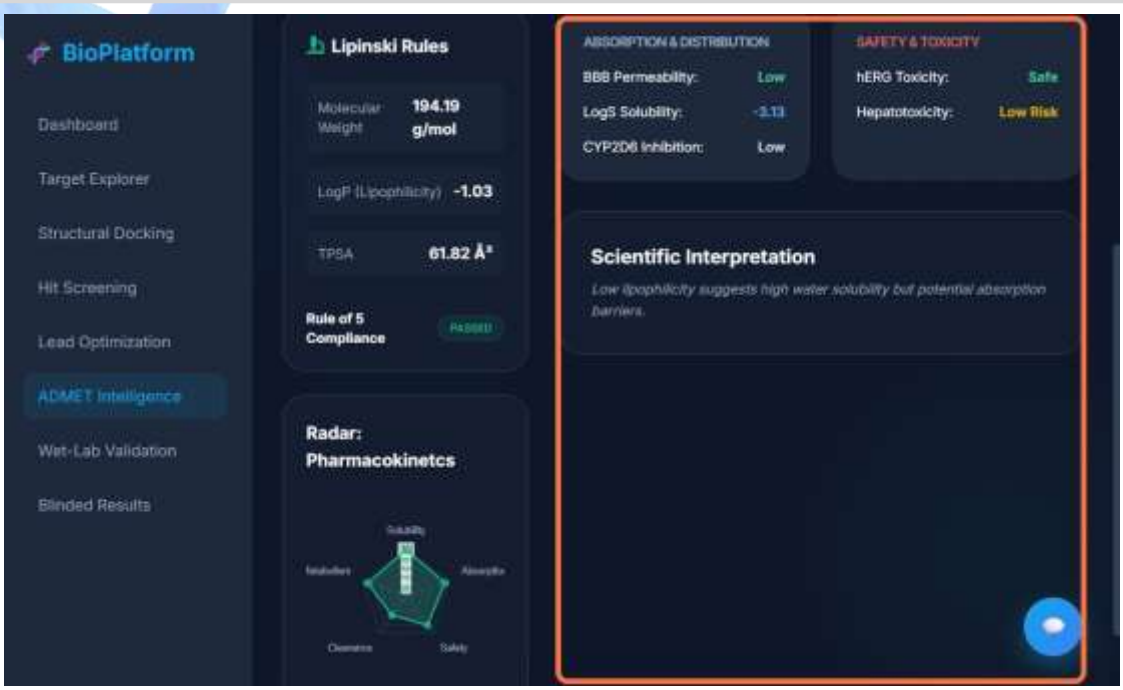
STEP 32:

Click on Predict Profile



STEP 33:

Click on ABSORPTION & DISTRIBUTION…

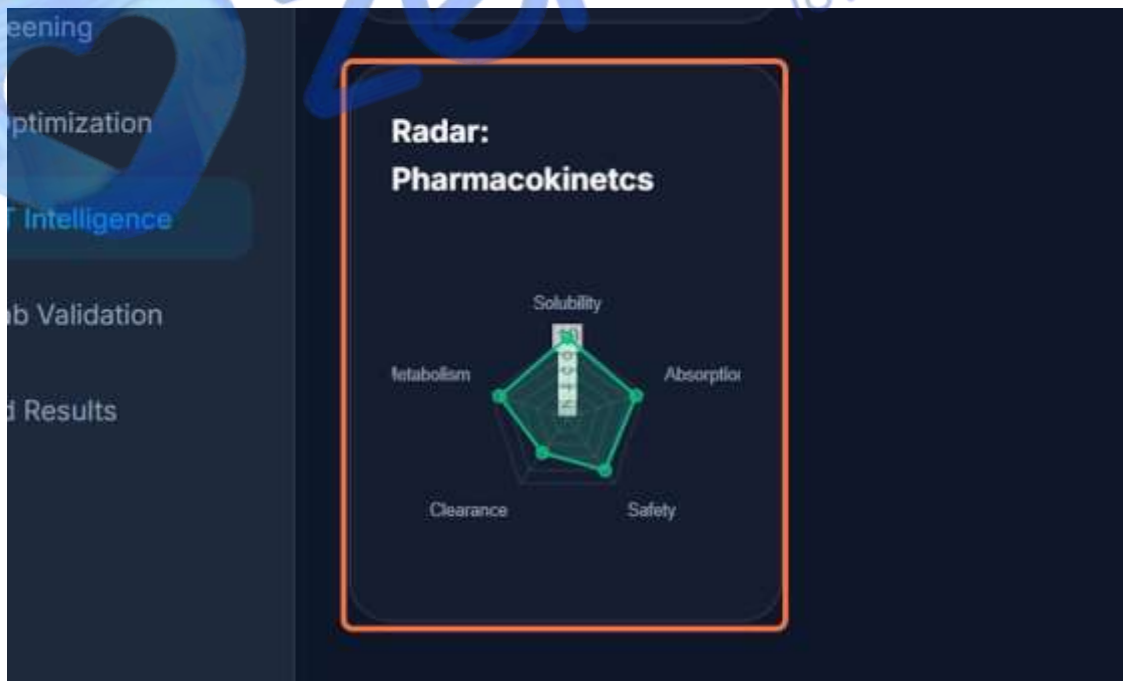


STEP 34:

Click on ADMET Intelligence…

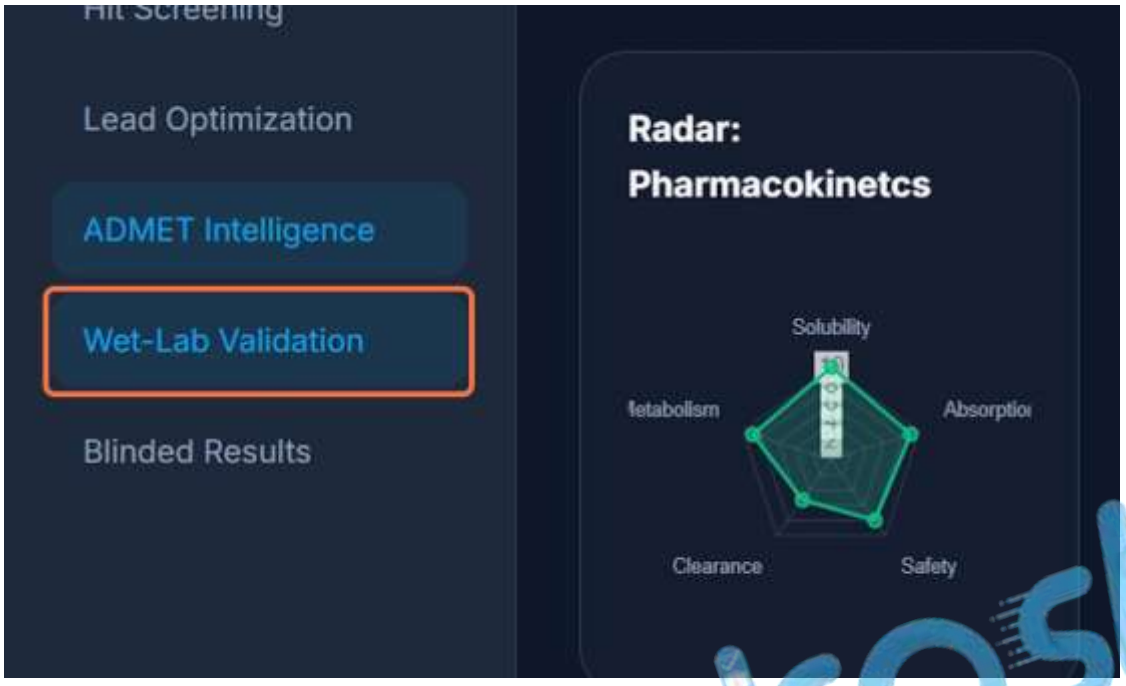
STEP 35:

Click on Radar: Pharmacokinetcs



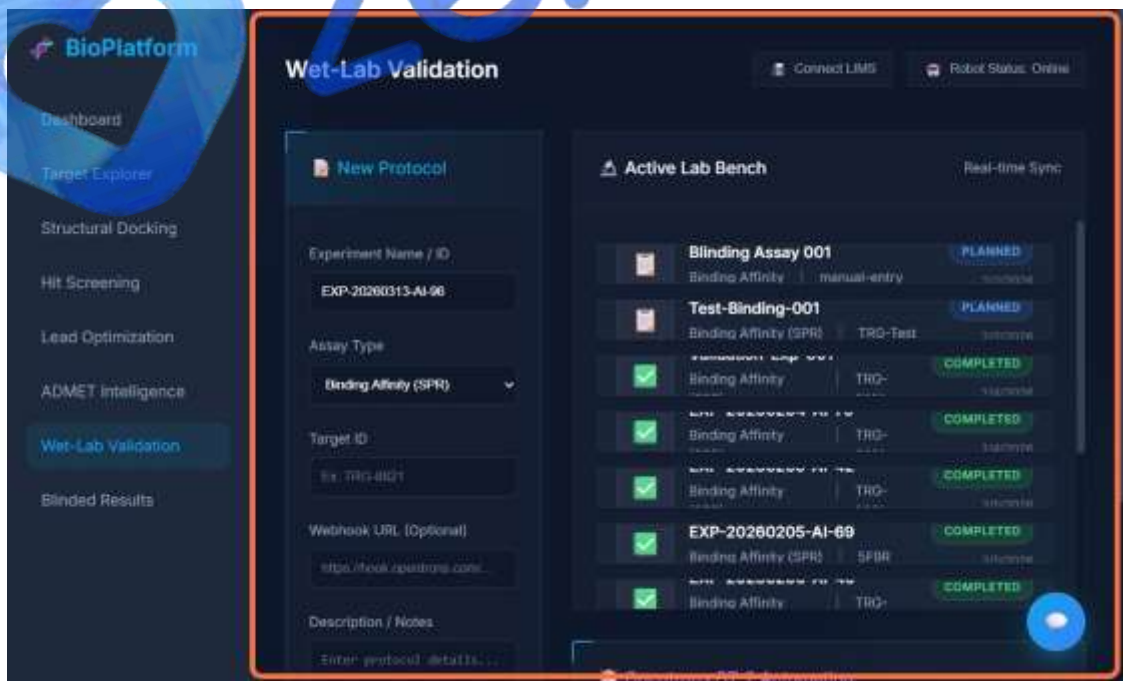
STEP 36:

Click on Wet-Lab Validation



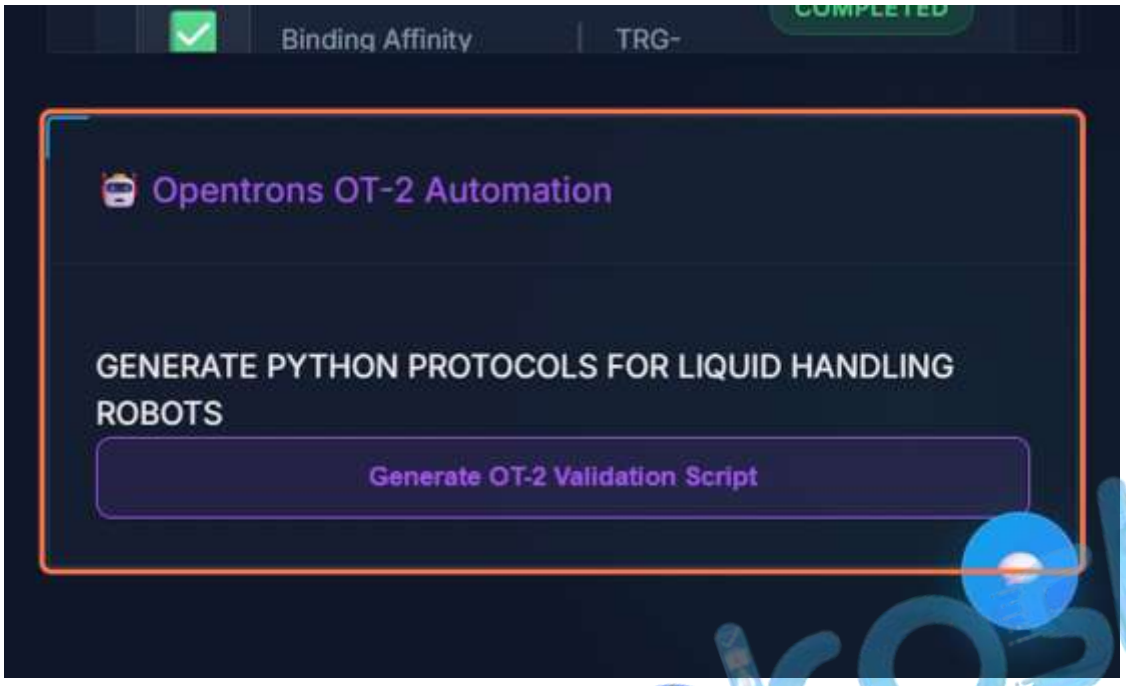
STEP 37:

Click on Wet-Lab Validation…



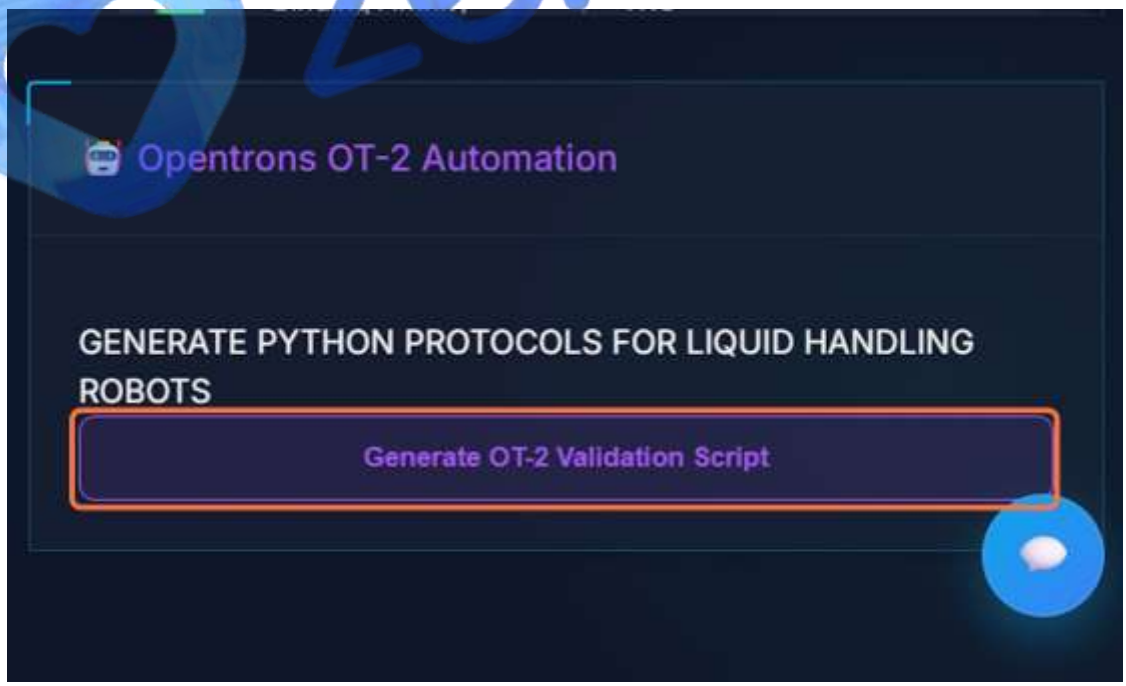
STEP 38:

Click on 🤖 Opentrons OT-2 Automation…



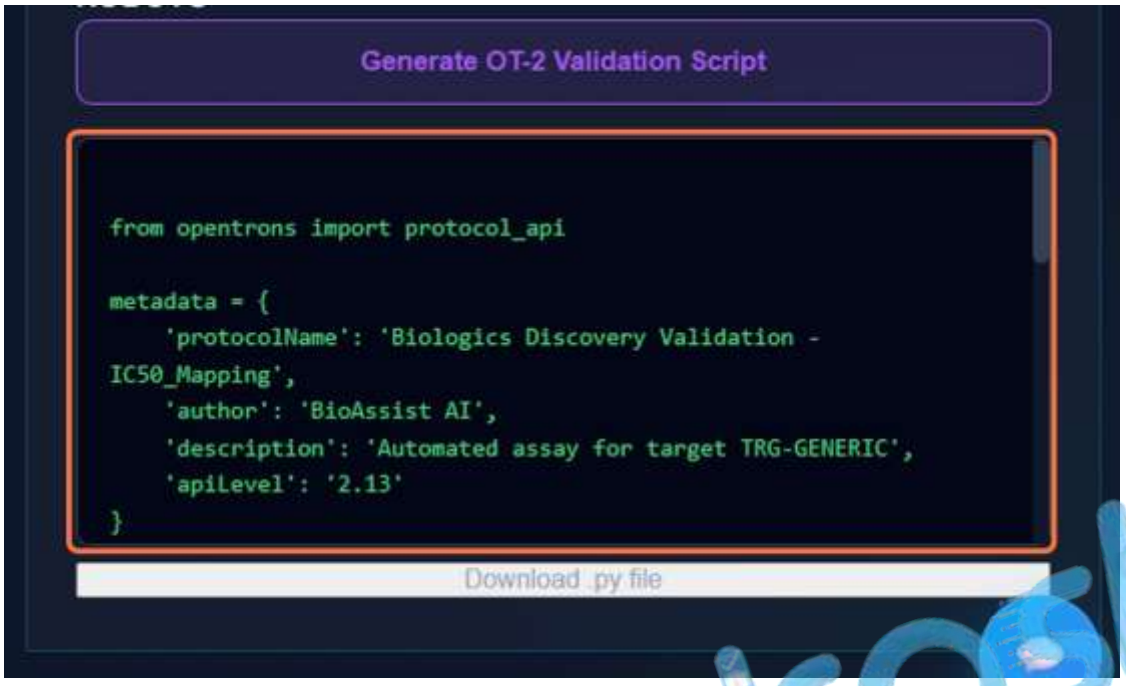
STEP 39:

Click on Generate OT-2 Validation Script



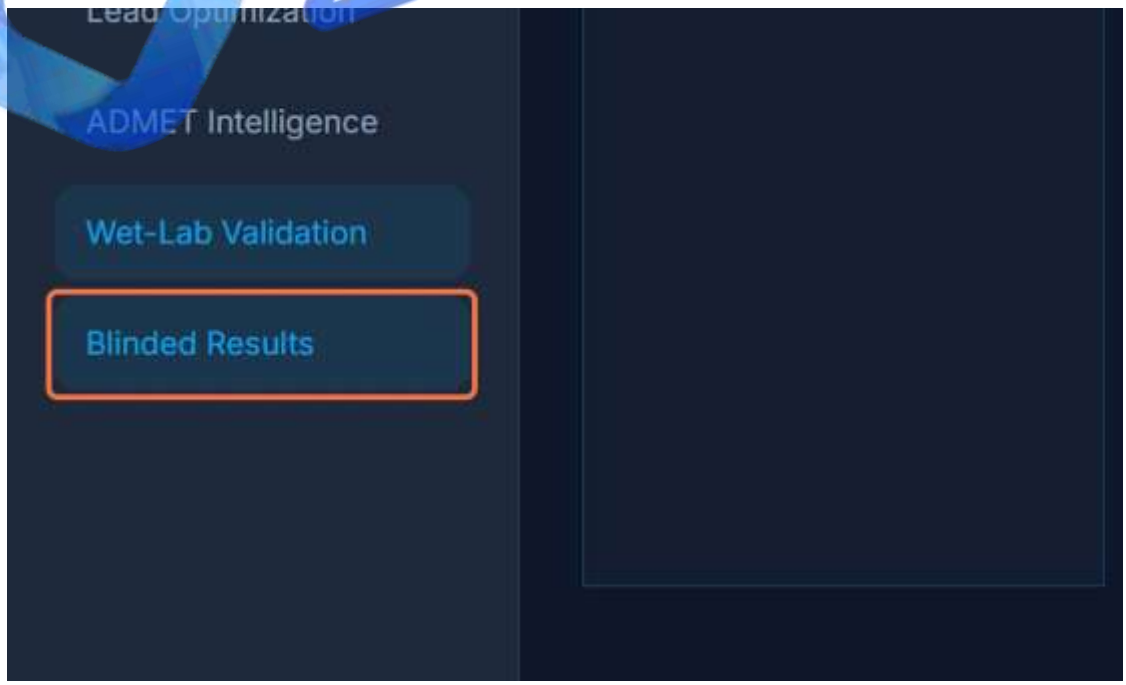
STEP 40:

Click on from opentrons import protocol_api…



STEP 41:

Click on Blinded Results

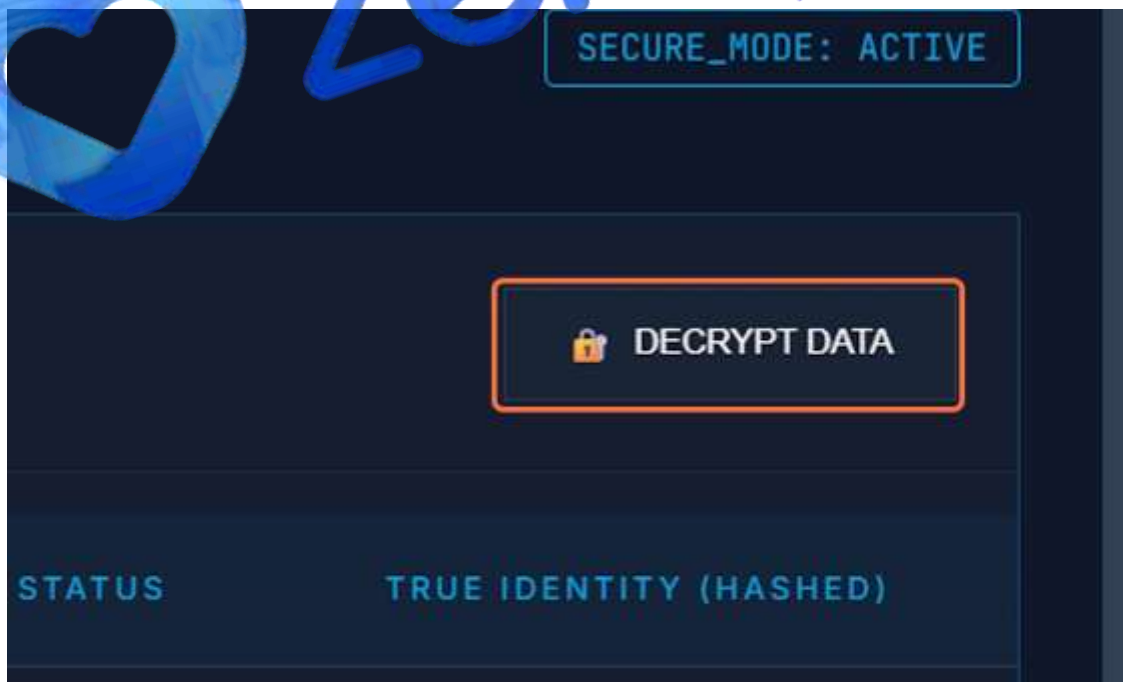


STEP 42:

Click on Blinded Study Design…

STEP 43:

Click on DECRYPT DATA

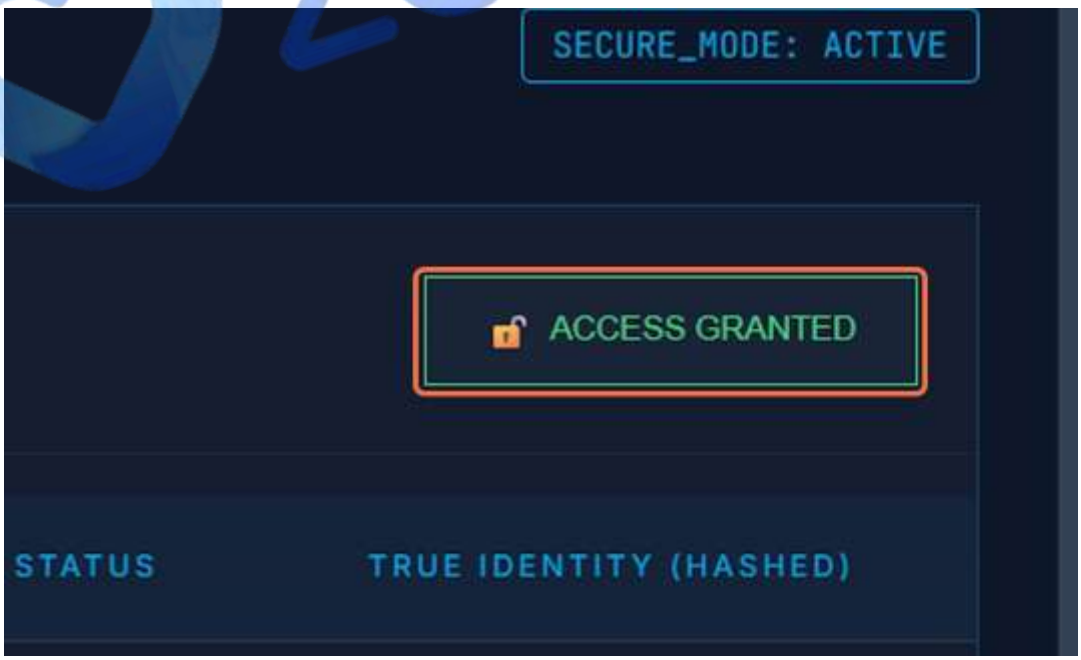


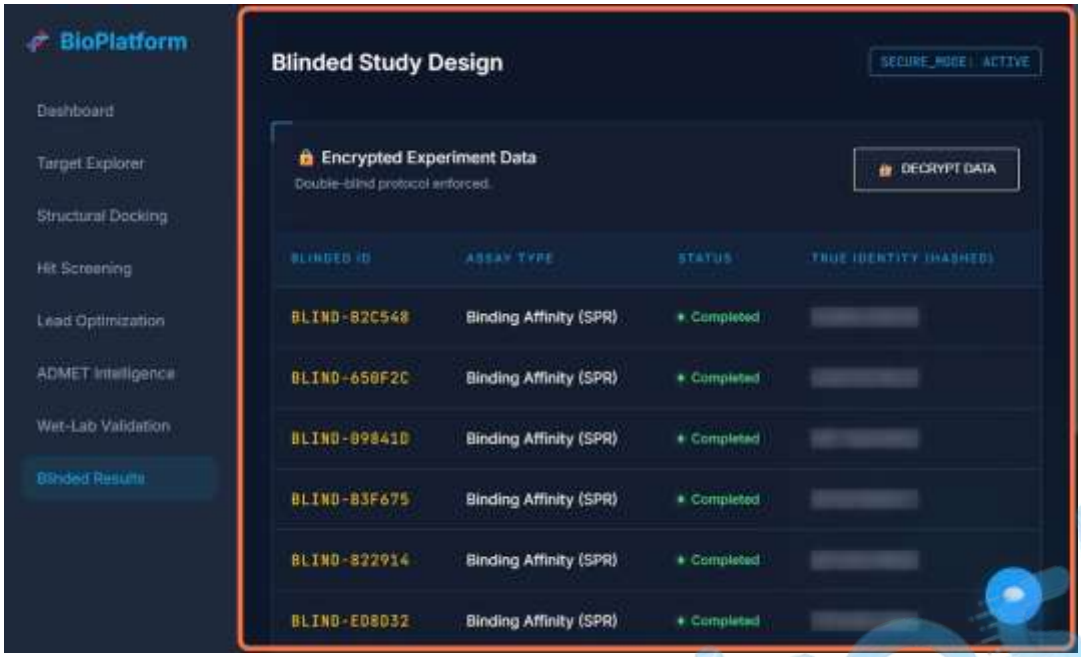
STEP 44:

Click on Blinded Study Design…

STEP 45:

Click on ACCESS GRANTED



STEP 46

Click on Blinded Study Design…